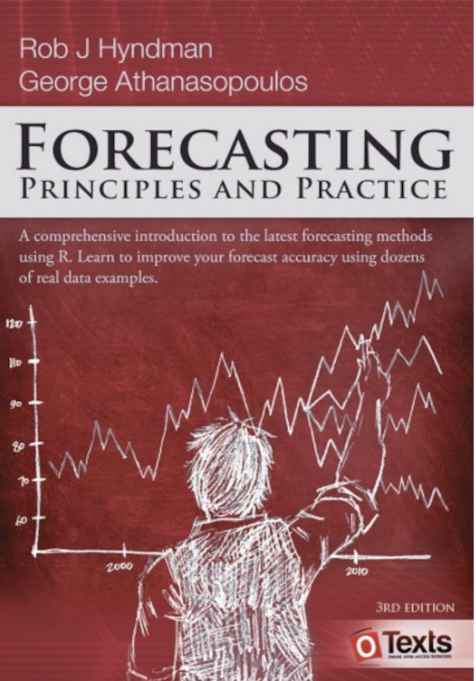


9. ARIMA models

9.6 Estimation and order selection

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Maximum likelihood estimation

Having identified the model order, we need to estimate the parameters c, $\phi_1, \ldots, \phi_p$, $\theta_1, \ldots, \theta_q$.

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- MLE is very similar to least squares estimation obtained by minimizing
$$\sum_{t=1}^{T} e_t^2$$
- The `ARIMA()` function allows conditional least squares (CLS) or MLE estimation.
- Non-linear optimization must be used in either case.
- Different software will give different estimates.

Information criteria

Akaike's Information Criterion (AIC):

$$\text{AIC} = -2\log(L) + 2(p + q + k + 1),$$

where L is the likelihood of the data, $k = 1$ if $c \neq 0$ and $k = 0$ if $c = 0$.

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Good models are obtained by minimizing either the AIC, AICc or BIC. Our preference is to use the AICc.